Electrical Science Formula Sheet

$i = dq/dt$ $\quad Q = \int_0^t i dt$ $\quad v_{ab} = dw/dq$ $\quad p = dw/dt = vi$ $\quad w = \int_{to}^t P dt$ $\quad v = iR$ $\quad G = 1/R$ $\quad p = vi = i^2R = v^2/R$

$\sum_{n=1}^N i_n = 0$ $\quad \sum_{n=1}^N v_n = 0$ $\quad$ Series: $R_{eq} = R_1 + R_2 + \ldots + R_n$ $\quad$ Parallel: $1/R_{eq} = 1/R_1 + 1/R_2 + \ldots + 1/R_n$

$v_n = (R_n*v)/(R_1+R_2+\ldots+R_N)$ $\quad q = Cv$ $\quad C=\varepsilon A/d$ $\quad i = C\, dv/dt$ $\quad v = \frac{1}{C}\int_{to}^t i dt + v(to)$ $\quad w=Cv^2/2$

Parallel: $C_{eq} = C_1 + C_2 + \ldots + C_n$ $\quad$ Series: $1/C_{eq} = 1/C_1 + 1/C_2 + \ldots + 1/C_n$ $\quad v = L\, di/dt$ $\quad L = N^2\mu A/l$

$i = \frac{1}{L}\int_{to}^t v(t)dt + i(to)$ $\quad w=Li^2/2$ $\quad$ Series: $L_{eq} = L_1 + L_2 + \ldots + L_n$ $\quad$ Parallel: $1/L_{eq} = 1/L_1 + 1/L_2 + \ldots + 1/L_n$

Capacitor discharge: $v(t) = V_0e^{-t/\tau}$ $\quad \tau = RC$ $\quad$ Inductor discharge: $i(t) = I_0e^{-t/\tau}$ $\quad \tau = L/R$

Capacitor step response $v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau}$

Inductor step response $i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau}$

Complex Variables: $z = x + jy$ $\quad z = re^{j\theta}$ $\quad r = \sqrt{x^2 + y^2}$ $\quad \theta = tan^{-1}(\frac{y}{x})$ $\quad z^* = x - jy$ $\quad z^* = re^{-j\theta}$

$z_1z_2 = r_1 r_2e^{j(\theta_1+\theta_2)}$ $\quad z_1/z_2 = (r_1/r_2)e^{j(\theta_1-\theta_1)}$ $\quad$ impedance: $Z_C = 1/j\omega C$ $\quad Z_L = j\omega L$ $\quad Z_R = R$

Steps to determine the node voltages:

1. Select a node as the reference node.
2. Assign voltages $v_1, v_2, \ldots, v_{n-1}$ to the remaining n-1 nodes. The voltages are referenced with respect to the reference node.
3. Apply KCL to each of the n-1 non-reference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
4. Solve the resulting simultaneous equations to obtain the unknown node voltages.

Steps to determine the node voltages with voltage source between nodes:

1. A super-node is formed by enclosing a (dependent or independent) voltage source connected between two non-reference nodes and any elements connected in parallel with it.
2. Take off all voltage sources in super-nodes and apply KCL to super-nodes.
3. Put voltage sources back to the nodes and apply KVL to relative loops.

Steps to determine the mesh currents:

1. Assign mesh currents $i_1, i_2, \ldots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

Steps to determine the mesh currents with current source:

A **super-mesh** results when two meshes have a (dependent or independent) current source in common.
1- We create a super-mesh by excluding the current source and any elements connected in series with it.
2- Write KVL in the super-mesh using the loop currents of the two adjacent meshes.
3- The sum or the difference of the two mesh currents equals the current source in the common branch between the two meshes.

Thevenin's Theorem

It states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{TH} in series with a resistor R_{TH},

- V_{TH} is the open-circuit voltage at the terminals.
- R_{TH} is the input or equivalent resistance at the terminals when the independent sources are turned off.
- Independent sources are turned off by replacing a Voltage source by a short circuit and current source by open circuit.

Norton’s Theorem

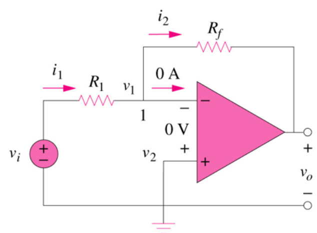
It states that a linear two-terminal circuit can be replaced by an equivalent circuit of a current source I_N in parallel with a resistor R_N.

- I_N is the short circuit current through the terminals.
- R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.
- Independent sources are turned off by replacing a Voltage source by a short circuit and current source by open circuit.

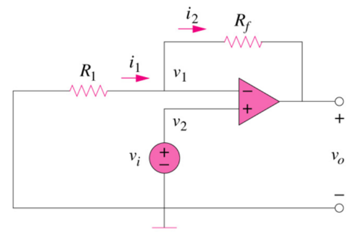
Maximum Power Transfer

For maximum Power transfer load resistor should equal Thevenin’s resistor

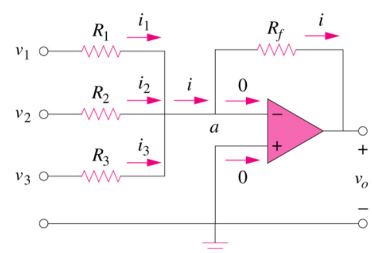
$$R_L = R_{TH} \quad \Rightarrow \quad P_{\max} = \frac{{V_{Th}}^2}{4R_L}$$



$$v_o = -\frac{R_f}{R_1} v_i$$

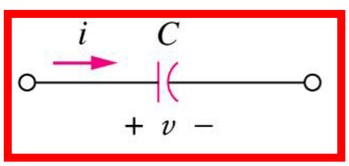


$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$



$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3\right)$$

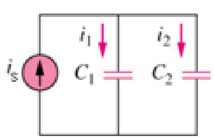
Capacitor



$$i = C \frac{dv}{dt}$$

$$v = \frac{1}{C} \int_{t_0}^{t} i \, dt + v(t_0)$$

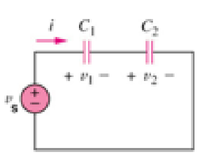
$$w = \frac{1}{2} C v^2$$



$$i_1 = \frac{C_1}{C_1 + C_2} i_s$$

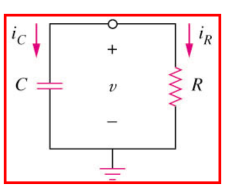
$$i_2 = \frac{C_2}{C_1 + C_2} i_s$$

$$C = \frac{\varepsilon A}{d} \qquad C = q/v$$

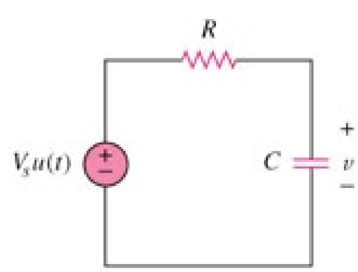


$$v_1 = \frac{C_2}{C_1 + C_2} v_s$$

$$v_2 = \frac{C_1}{C_1 + C_2} v_s$$

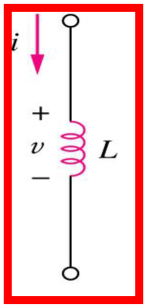


$$v(t) = V_0 \, e^{-t/\tau}$$ where $$\tau = RC$$

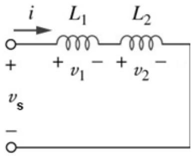


$$v(t) = \begin{cases} V_0 & t < 0 \\ V_s + (V_0 - V_s) e^{-t/\tau} & t > 0 \end{cases}$$

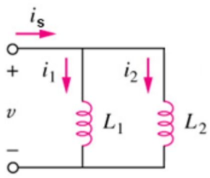
Inductor



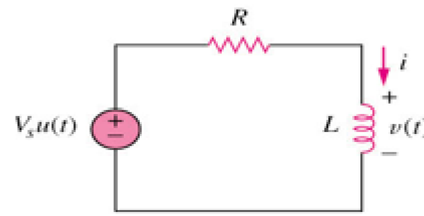
$$v = L\frac{di}{dt}$$ $$L = \frac{N^2 \mu A}{l}$$ $$i = \frac{1}{L}\int_{t_0}^{t} v(t)\,dt + i(t_0)$$ $$w = \frac{1}{2}Li^2$$

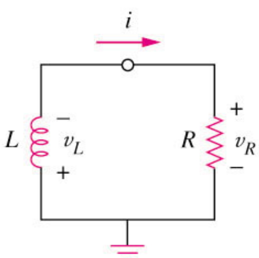


$$v_1 = \frac{L_1}{L_1 + L_2}v_s$$ $$v_2 = \frac{L_2}{L_1 + L_2}v_s$$



$$i_1 = \frac{L_2}{L_1 + L_2}i_s$$ $$i_2 = \frac{L_1}{L_1 + L_2}i_s$$





$$i(t) = I_0\, e^{-t/\tau}$$ where $$\tau = \frac{L}{R}$$

$$i(t) = \frac{V_s}{R} + (I_o - \frac{V_s}{R})e^{-\frac{t}{\tau}}u(t)$$

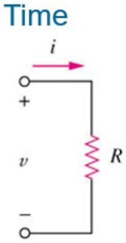
$$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow V = V_m \angle\phi$$

(time domain) (phasor domain)

Addition	$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$
Subtraction	$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$
Multiplication	$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$
Division	$\frac{z_1}{z_2} = \frac{r_1}{r_2}\angle \phi_1 - \phi_2$
Reciprocal	$\frac{1}{z} = \frac{1}{r}\angle -\phi$
Square root	$\sqrt{z} = \sqrt{r}\angle \phi/2$
Complex conjugate	$z^* = x - jy = r\angle -\phi = re^{-j\phi}$
Euler's identity	$e^{\pm j\phi} = \cos\phi \pm j\sin\phi$

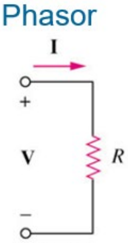
Resistor: Inductor: Capacitor:

Time



$v = iR$

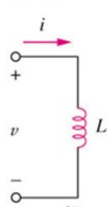
Phasor



$\mathbf{V} = \mathbf{I}R$

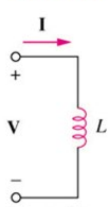
Z = R

Time



$v = L\frac{di}{dt}$

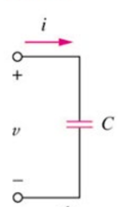
Phasor



$\mathbf{V} = j\omega L\mathbf{I}$

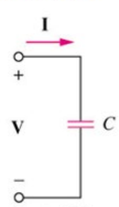
Z = $j\omega L$

Time



$i = C\frac{dv}{dt}$

Phasor



$\mathbf{I} = j\omega C\mathbf{V}$

Z = $1/j\omega C$